

CSE 432 Robotics

Mobile Robots II

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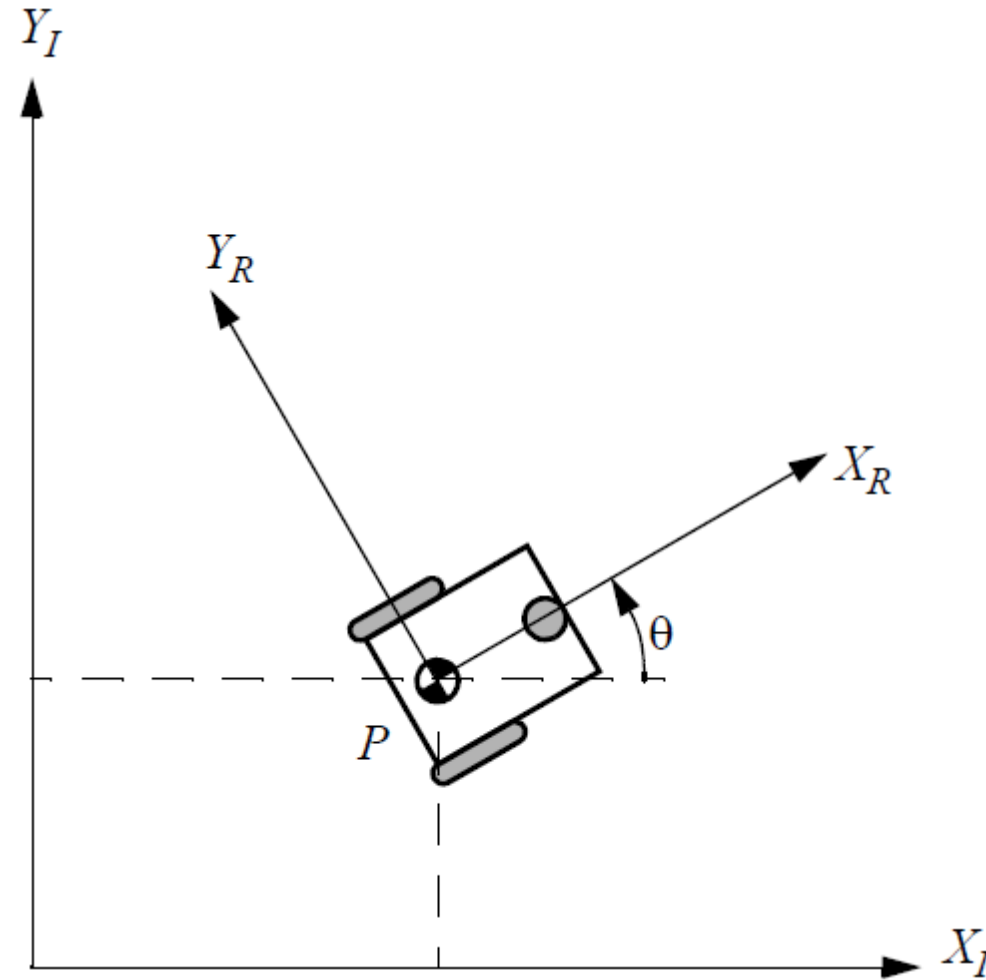


Kinematic Models and Constraints

- Wheels are arranged based on robot chassis geometry.
- Thus, their constraints combine to form constraints on the overall motion of the robot chassis.
- The forces and constraints of each wheel must be expressed with respect to a consistent reference frame.
- A clear mapping between global and local frames of reference is required.



Representing robot position





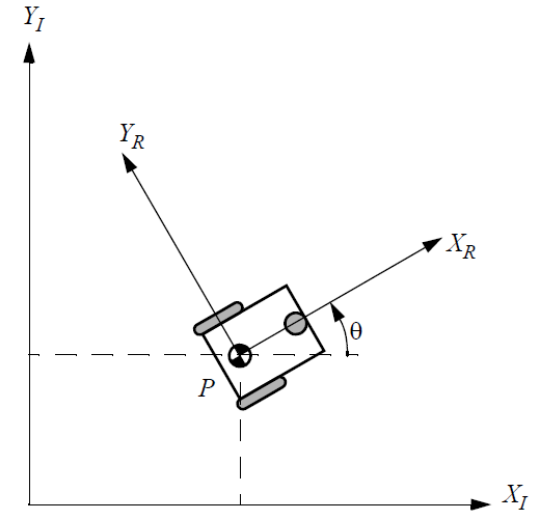
Representing robot position

- The pose of the robot, ξ :

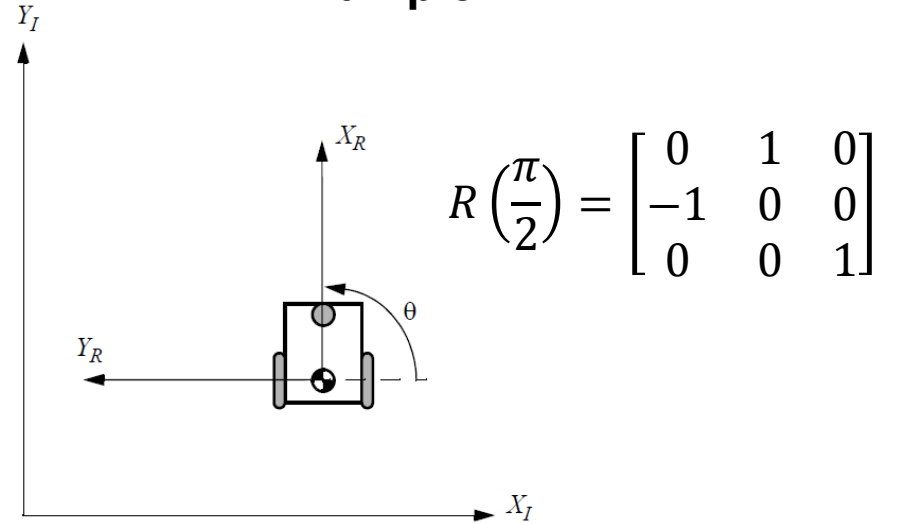
$$\xi = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$$

- R can be used to map motion in the global reference, ξ_I frame to motion in terms of the local reference frame ξ_R .

$$R = \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Example





Representing robot position

- The matrix R can be used to map motion in the global reference frame to motion in terms of the local reference frame.

$$\dot{\xi}_R = R(\theta)\dot{\xi}_I$$

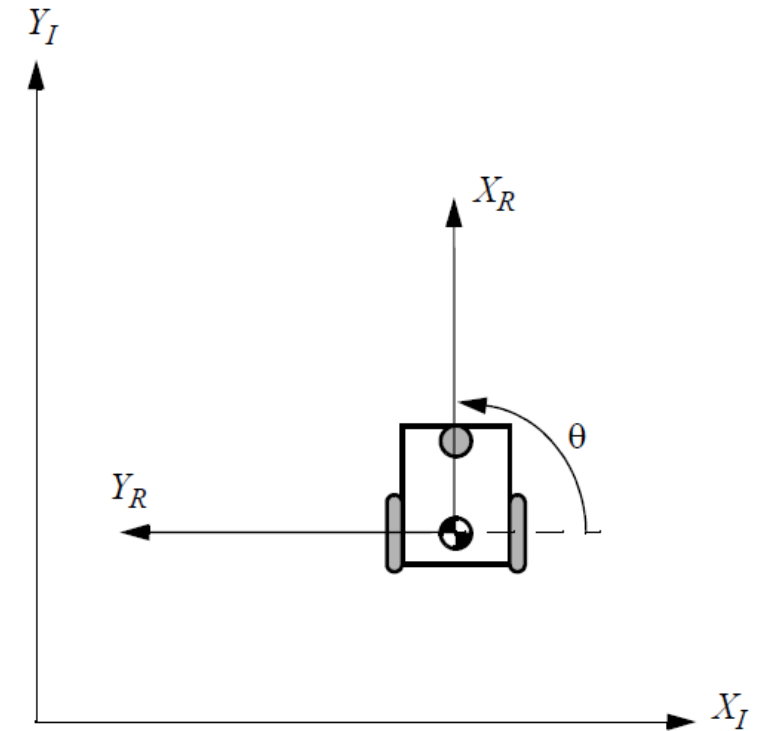
- Given some velocity $(\dot{x}, \dot{y}, \dot{\theta})$ in the global reference frame we can compute the components of motion along this robot's local axes X_R and Y_R .



Representing robot position

- Example

$$\dot{\xi}_R = R\left(\frac{\pi}{2}\right) \dot{\xi}_I = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \dot{y} \\ -\dot{x} \\ \dot{\theta} \end{bmatrix}$$

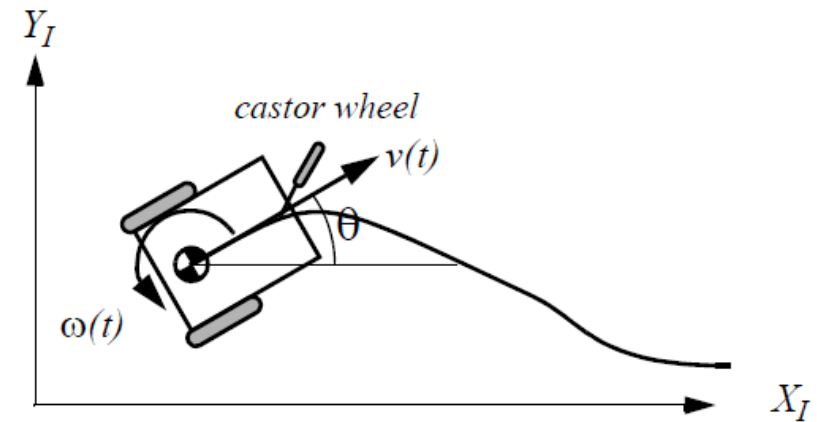




Forward kinematic models

- This differential drive robot has two wheels, each with diameter r . Given a point P centered between the two drive wheels, each wheel is a distance l from P .
- Given r, l, θ and the spinning speed of each wheel, $\dot{\phi}_1$ and $\dot{\phi}_2$, a forward kinematic model would predict the robot's overall speed in the global reference frame:

$$\dot{\xi}_I = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = f(l, r, \theta, \dot{\phi}_1, \dot{\phi}_2)$$





Wheel kinematic constraints

- The first step to a kinematic model of the robot is to express constraints on the motions of individual wheels.
- Individual wheel motions can be combined to get overall robot motion.
- There are four basic wheel types with widely varying kinematic properties.

Assumption in Kinematic Modelling:

- The plane of the wheel always remains vertical and that there is in all cases one single point of contact between the wheel and the ground plane.
- There is no sliding at this single point of contact.
- The wheel undergoes motion only under conditions of pure rolling and rotation about the vertical axis through the contact point.



Wheel kinematic constraints

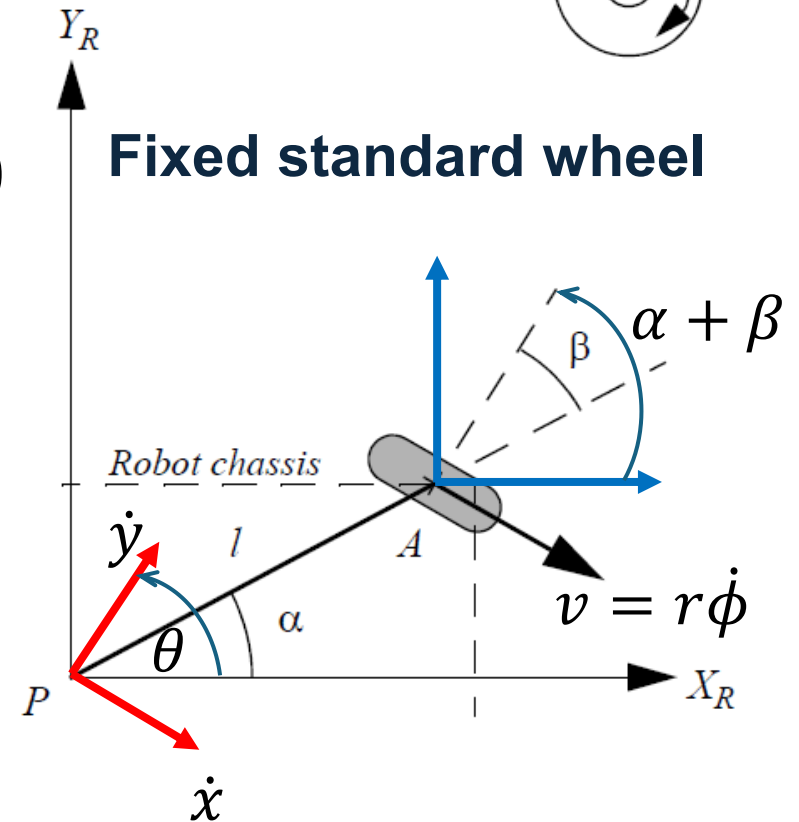
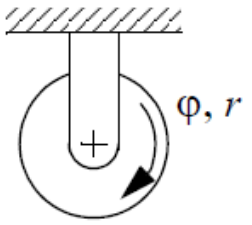
- **Rolling Constraint: Fixed standard wheel**

$$[\sin(\alpha + \beta) \quad -\cos(\alpha + \beta) \quad -l \cos \beta] R(\theta) \dot{\xi}_I - r \dot{\phi} = 0$$

- **Sliding Constraint: Fixed standard wheel**

$$[\cos(\alpha + \beta) \quad \sin(\alpha + \beta) \quad l \sin \beta] R(\theta) \dot{\xi}_I = 0$$

- β : The angle of the wheel plane relative to the chassis.
- The position of A is expressed in polar coordinates by distance l and angle β .





Wheel kinematic constraints

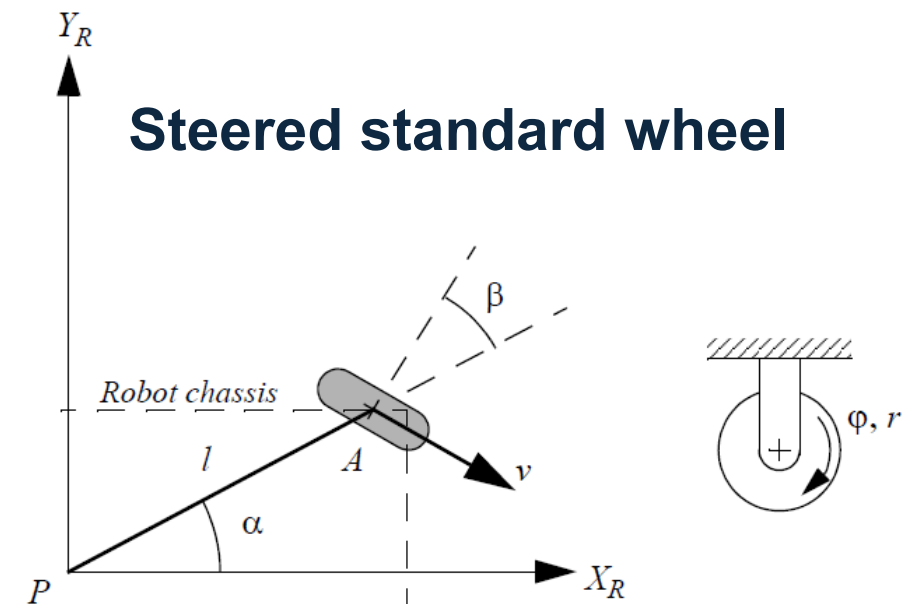
- **Rolling Constraint: Steered standard wheel**

$$[\sin(\alpha + \beta) \quad -\cos(\alpha + \beta) \quad -l \cos \beta] R(\theta) \dot{\xi}_I - r \dot{\phi} = 0$$

- **Sliding Constraint: Steered standard wheel**

$$[\cos(\alpha + \beta) \quad \sin(\alpha + \beta) \quad l \sin \beta] R(\theta) \dot{\xi}_I = 0$$

- $\dot{\phi}$ and $\dot{\beta}$ does not have a direct impact on the instantaneous motion constraints of a robot. It is only by integrating over time that changes in steering angle can affect the mobility of a vehicle.
- Same as fixed standard wheel but β varies as a function of time $\beta(t)$





Wheel kinematic constraints

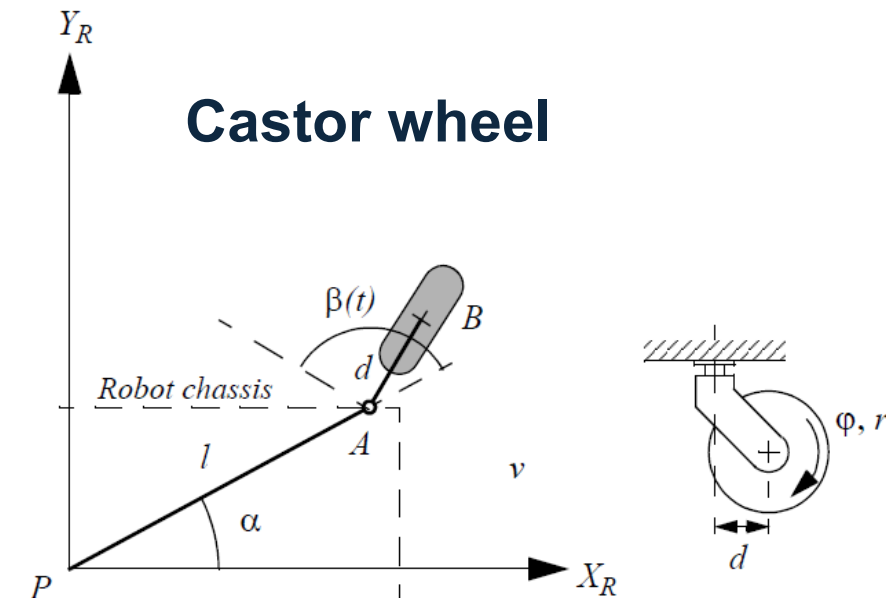
- **Rolling Constraint: Castor wheel**

$$[\sin(\alpha + \beta) \quad -\cos(\alpha + \beta) \quad -l \cos \beta] R(\theta) \dot{\xi}_I - r \dot{\phi} = 0$$

- **Sliding Constraint: Castor wheel**

$$[\cos(\alpha + \beta) \quad \sin(\alpha + \beta) \quad d + l \sin \beta] R(\theta) \dot{\xi}_I + d \dot{\beta} = 0$$

The wheel contact point is now at position B, which is connected by a rigid rod AB of fixed length d to point A, fixes the location of the vertical axis about which B steers, and this point A has a position specified in the robot's reference frame.





Wheel kinematic constraints

- For the castor wheel, the rolling constraint is identical to fixed standard wheel because the offset axis plays no role during motion that is aligned with the wheel plane.
- The castor geometry does have significant impact on the sliding constraint.
- The critical issue is that the lateral force on the wheel occurs at point A because this is the attachment point of the wheel to the chassis.
- Any motion orthogonal to the wheel plane must be balanced by an equivalent and opposite amount of castor steering motion.
- By setting $\dot{\beta}$ any arbitrary lateral motion can be acceptable.



Wheel kinematic constraints

- Thus, given *any* robot chassis motion , there exists some value for spin speed $\dot{\phi}$ and steering speed $\dot{\beta}$ such that the constraints are met.
- Therefore, a robot with only castor wheels can move with any velocity in the space of possible robot motions.
- We term such systems *omnidirectional*.
- A real-world example of such a system is the five-castor wheel office chair
- Although the kinematics of castor wheels is somewhat complex, such wheels do not impose any real constraints on the kinematics of a robot chassis.





Wheel kinematic constraints

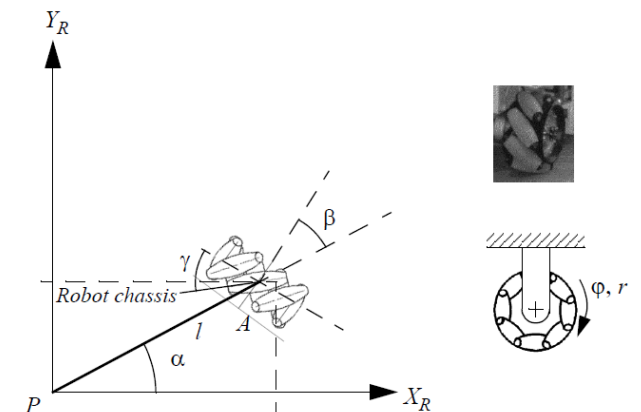
- **Rolling Constraint: Swedish wheel**

$$[\sin(\alpha + \beta + \gamma) \quad -\cos(\alpha + \beta + \gamma) \quad -l \cos(\beta + \gamma)]R(\theta)\dot{\xi}_I - r\dot{\phi} \cos \gamma = 0$$

- Orthogonal to this direction the motion is not constrained because of the free rotation of the small rollers $\dot{\phi}_{sw}$.

$$[\cos(\alpha + \beta + \gamma) \quad \sin(\alpha + \beta + \gamma) \quad l \sin(\beta + \gamma)]R(\theta)\dot{\xi}_I - r\dot{\phi} \sin \gamma - \dot{r}_{sw}\dot{\phi}_{sw} = 0$$

That matches the fixed standard wheel's rolling constraint, but with an added γ term that shifts the constraint direction to the **zero component**, not the wheel plane.





Wheel kinematic constraints

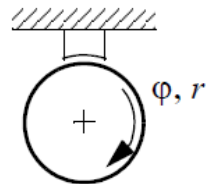
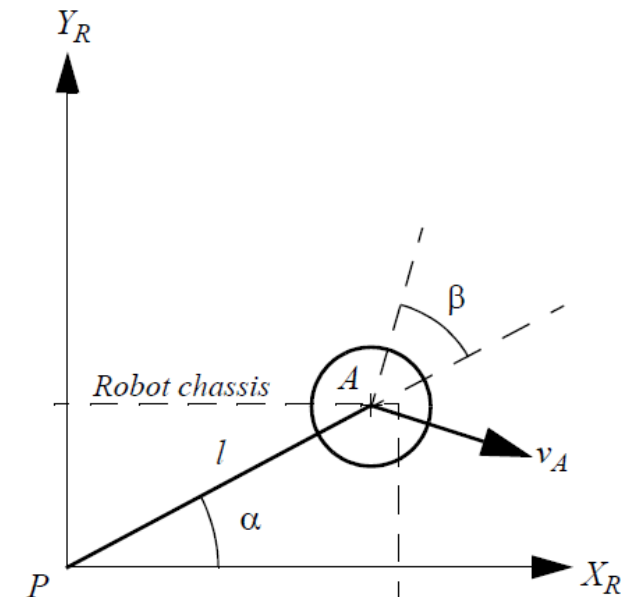
- **Rolling Constraint: Spherical wheel**

$$[\sin(\alpha + \beta) \quad -\cos(\alpha + \beta) \quad -l \cos \beta] R(\theta) \dot{\xi}_I - r \dot{\phi} = 0$$

- The wheel rotation orthogonal to the rolling direction is zero

$$[\cos(\alpha + \beta) \quad \sin(\alpha + \beta) \quad l \sin \beta] R(\theta) \dot{\xi}_I = 0$$

- A ball or spherical wheel, places no direct constraints on motion.
- Such a mechanism has no principal axis of rotation, and therefore no appropriate rolling or sliding constraints exist.





Wheel kinematic constraints

- Like castor and Swedish wheels, the spherical wheel is inherently omnidirectional and does not constrain the chassis kinematics.
- Although the equations describing the spherical wheel are the same as those for the fixed standard wheel, their physical interpretation is different.
- The omnidirectional spherical wheel can have any arbitrary direction of movement, where the motion direction given by is a free variable.



Robot kinematic constraints

- Given a mobile robot with M wheels we can compute the kinematic constraints of the robot chassis.
- The key idea is that each wheel imposes zero or more constraints on robot motion, and so the process is simply
 - Combining all kinematic constraints imposed by the wheels, based on their placement on the robot chassis.
- We have categorized all wheels into five categories:
 - (1) fixed standard wheels,
 - (2) steerable standard wheels,
 - (3) castor wheels,
 - (4) Swedish wheels,
 - (5) spherical wheels.



Robot kinematic constraints

- Note from the wheel kinematic constraints that the castor wheel, Swedish wheel, and spherical wheel impose *no sliding* kinematic constraints on the robot chassis, since $\dot{\xi}_I$ can range freely in all of these cases owing to the internal wheel degrees of freedom.
- Only **fixed standard** wheels and **steerable standard wheels** have impact on robot chassis kinematics and therefore require consideration when computing the robot's kinematic constraints.



Robot kinematic constraints

- Suppose that the robot has a total of N standard wheels, comprising N_f fixed standard wheels and N_s steerable standard wheels.
- We use $\dot{\beta}_s(t)$ to denote the variable steering angles of the steerable standard wheels.
- In the case of wheel spin, both the fixed and steerable wheels have rotational positions around the horizontal axle that vary as a function of time (spinning speed).



Robot kinematic constraints

- Denote the fixed and steerable cases as $\phi_f(t)$ and $\phi_s(t)$, and
- Use $\phi(t)$ as the aggregate matrix combining both:

$$\phi(t) = \begin{bmatrix} \phi_f(t) \\ \phi_s(t) \end{bmatrix}$$

- The rolling constraints of all wheels: $J_1(\beta_s)R(\theta)\dot{\xi}_I - J_2\dot{\phi} = 0$
- $J_2 \in \mathbb{R}^{N \times N}$ is a diagonal matrix containing the standard wheel radii r .
- $J_1(\beta_s)$ denotes a matrix with projections for all wheels to their motions along their individual wheel planes:

$$J_1(\beta_s) = \begin{bmatrix} J_{1f} \\ J_{1s}(\beta_s) \end{bmatrix}$$



Robot kinematic constraints

- Note that $J_1(\beta_s)$ is only a function of β_s and not β_f .
- J_{1f} is therefore a constant matrix of projections for all fixed standard wheels. It has size $(N_f \times 3)$, with each row consisting of the three terms in the three matrix for each fixed standard wheel.
- J_{1s} is a matrix of size $(N_s \times 3)$, with each row consisting of the three terms in the three-matrix for each steerable standard wheel.
- This constraint ensures that each standard wheel rotates about its horizontal axis by the correct amount relative to its motion in the wheel plane, so that pure rolling occurs at the ground contact point.



Robot kinematic constraints

- We use the same technique to collect the sliding constraints of all standard wheels into a single expression with the same structure as follows:

$$C_1(\beta_s)R(\theta)\dot{\xi}_I = 0,$$
$$C_1(\beta_s) = \begin{bmatrix} C_{1f} \\ C_{1s}(\beta_s) \end{bmatrix}$$

- C_{1f} and C_{1s} are $(N_f \times 3)$ and $(N_s \times 3)$ matrices whose rows are the three terms in the three-matrix of equations for all fixed and steerable standard wheels.
- This constraint ensures each standard wheel rotates about its horizontal axis in proportion to its motion in the wheel plane, enabling pure rolling at the ground contact point.

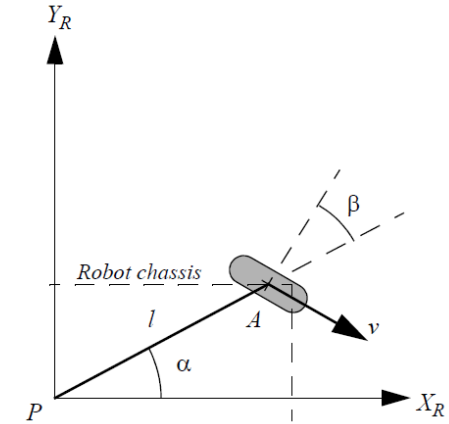


Differential-drive Robot

- The rolling & Sliding constraints :

$$J_1(\beta_s)R(\theta)\dot{\xi}_I - J_2\dot{\phi} = 0$$

$$C_1(\beta_s)R(\theta)\dot{\xi}_I = 0$$



- Fusing these two equations yields the following expression:

$$\begin{bmatrix} J_1(\beta_s) \\ C_1(\beta_s) \end{bmatrix} R(\theta)\dot{\xi}_I = \begin{bmatrix} J_2\dot{\phi} \\ 0 \end{bmatrix}$$

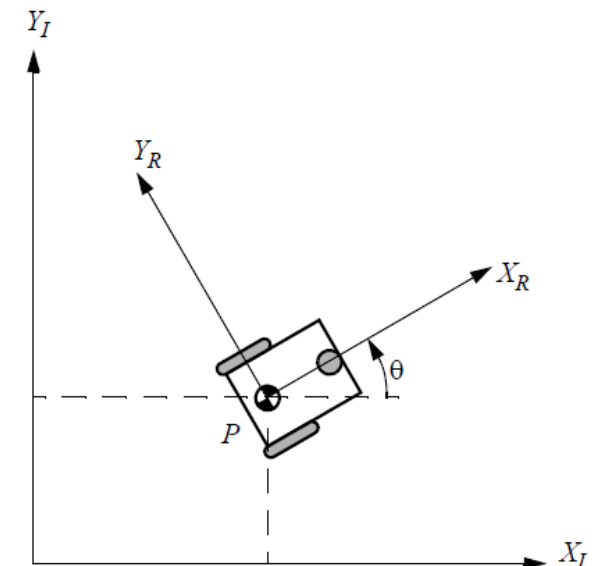
- Recalling that for standard fixed wheel

- Rolling Constraint

$$[\sin(\alpha + \beta) \quad -\cos(\alpha + \beta) \quad -l \cos \beta] R(\theta)\dot{\xi}_I - r\dot{\phi} = 0$$

- Sliding Constraint

$$[\cos(\alpha + \beta) \quad \sin(\alpha + \beta) \quad l \sin \beta] R(\theta)\dot{\xi}_I = 0$$





Differential-drive Robot

- From robot geometry

$\alpha_R = -\frac{\pi}{2}, (\beta_R = \pi$ Assuming +ve rotation move the robot forward)

$$\alpha_L = \frac{\pi}{2}, \beta_L = 0$$

- Thus,

$$J_1 = \begin{bmatrix} 1 & 0 & l \\ 1 & 0 & -l \end{bmatrix}, J_2 = \begin{bmatrix} r_R & 0 \\ 0 & r_L \end{bmatrix}$$

- The two fixed standard wheels are parallel, thus, sliding constrains results is not independent, which yields one equation. Hence:

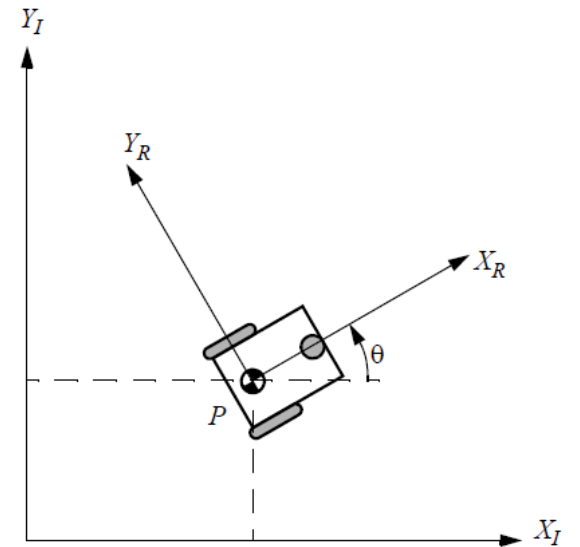
$$\begin{bmatrix} \begin{bmatrix} 1 & 0 & l \\ 1 & 0 & -l \end{bmatrix} \\ \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \end{bmatrix} R(\theta) \dot{\xi}_I = \begin{bmatrix} J_2 \phi \\ \mathbf{0} \end{bmatrix}$$



Differential-drive Robot

- Inverting the previous equations yields the kinematic equation specific to our differential drive robot:

$$\dot{\xi}_I = R(\theta)^{-1} \begin{bmatrix} \begin{bmatrix} 1 & 0 & l \\ 1 & 0 & -l \\ 0 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} J_2 \phi \\ 0 \end{bmatrix} \\ \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \\ \frac{1}{2l} & -\frac{1}{2l} & 0 \end{bmatrix} \begin{bmatrix} r_R \dot{\phi}_r \\ r_L \dot{\phi}_l \\ 0 \end{bmatrix} \end{bmatrix}$$



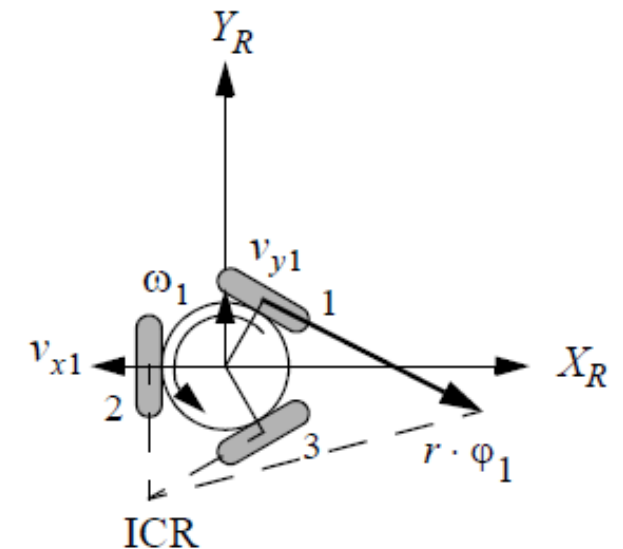


Omni-directional Robot

- The robot has three Swedish 90-degree wheels, arranged radially symmetrically, with the rollers perpendicular to each main wheel.
- The local reference frame is placed such that X_R is colinear with the axis of wheel 2.



A three-wheel omni-drive robot developed by Carnegie Mellon University.





Omni-directional Robot

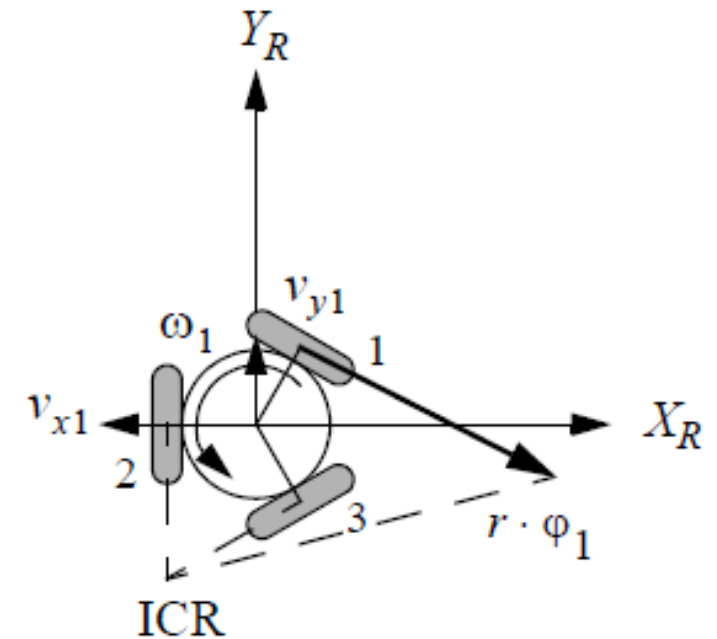
- We assume that the distance between each wheel and P is l , and that all three wheels have the same radius, r .
- $\dot{\xi}_I$ can be computed as a combination of the rolling constraints of the robot's three omnidirectional wheels.

$$\dot{\xi}_I = R(\theta)^{-1} J_1^{-1} J_2 \dot{\phi}$$

- For the Swedish 90-degree wheel.
- From the robot geometry

$$\alpha_1 = \frac{\pi}{3}, \alpha_2 = \pi, \alpha_3 = -\frac{\pi}{3},$$

$(\beta = 0 \text{ for all wheels})$





Omni-directional Robot

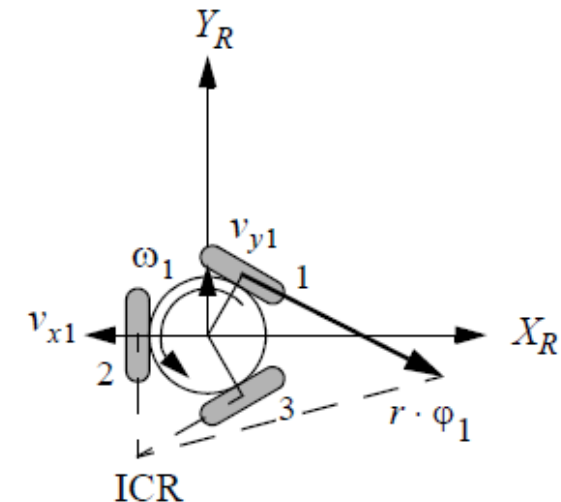
$$J_{1f} = \begin{bmatrix} \sin \frac{\pi}{3} & -\cos \frac{\pi}{3} & -l \\ 0 & -\cos \pi & -l \\ \sin -\frac{\pi}{3} & -\cos -\frac{\pi}{3} & -l \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & -l \\ 0 & 1 & -l \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} & -l \end{bmatrix}$$
$$\dot{\xi}_I = R(\theta)^{-1} \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{l}{\sqrt{3}} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{l}{3} \\ -\frac{1}{3l} & -\frac{1}{3l} & -\frac{l}{3l} \end{bmatrix} J_2 \dot{\phi}$$



Omni-directional Robot

- If $l = 1$ and $r = 1$ for all wheels.
- For $\theta = 0$, wheels 1, 2, and 3 spin at speeds ($\dot{\phi}_1 = 4$), ($\dot{\phi}_2 = 1$), ($\dot{\phi}_3 = 2$), the resulting motion of the whole robot can be calculated

$$\dot{\xi}_I = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & 0 & -\frac{1}{\sqrt{3}} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3l} & -\frac{1}{3l} & -\frac{1}{3l} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{3}} \\ \frac{4}{3} \\ -\frac{7}{3} \end{bmatrix}$$





Mobile Robot Maneuverability

- The basic constraint limiting mobility is the rule that every wheel must satisfy its sliding constraint.

$$C_{1f}R(\theta)\dot{\xi}_I = 0$$
$$C_{1s}(\beta_s)R(\theta)\dot{\xi}_I = 0$$

- For both of these constraints to be satisfied, the motion vector $R(\theta)\dot{\xi}_I$ must belong to the *null space* of the projection matrix $C_1(\beta_s)$, which is simply a combination of C_{1f} and C_{1s} .
- Mathematically, the null space of , $C_1(\beta_s)$ is the space N such that for any vector n in N , $C_1(\beta_s)n = 0$.
- The kinematic constraints can also be demonstrated geometrically using the concept of a robot's *instantaneous center of rotation* (ICR).



Mobile Robot Maneuverability

- A single standard wheel is subject to a **sliding constraint** that enforces **zero lateral motion**.
- Geometrically, this constraint is represented by a **zero-motion line**:
 - Drawn through the wheel center
 - Perpendicular to the wheel plane (aligned with the wheel's horizontal axis)
- At any instant, the wheel's velocity **along the zero-motion line is zero**.

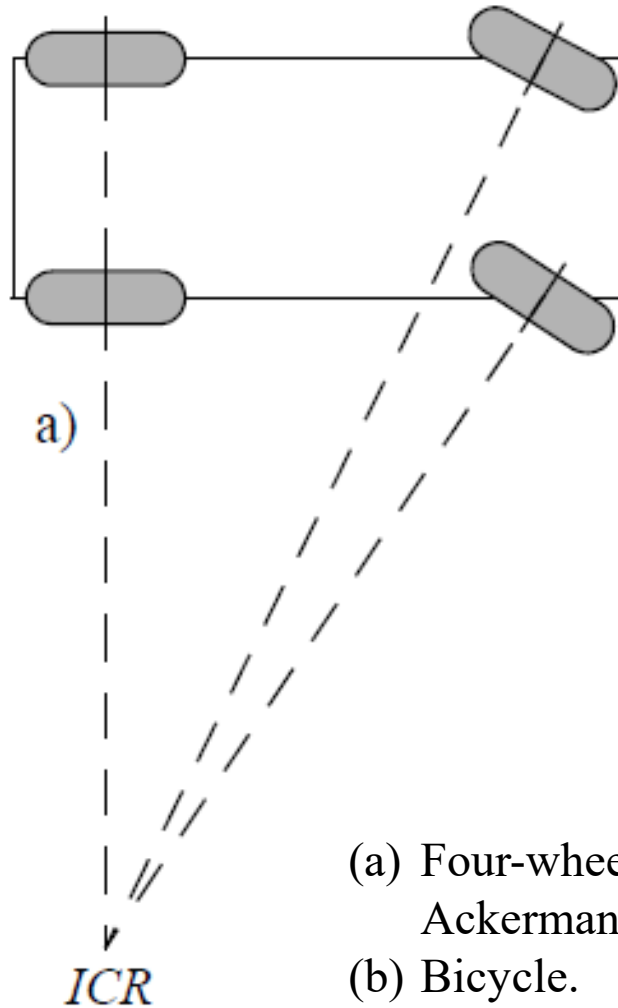


Mobile Robot Maneuverability

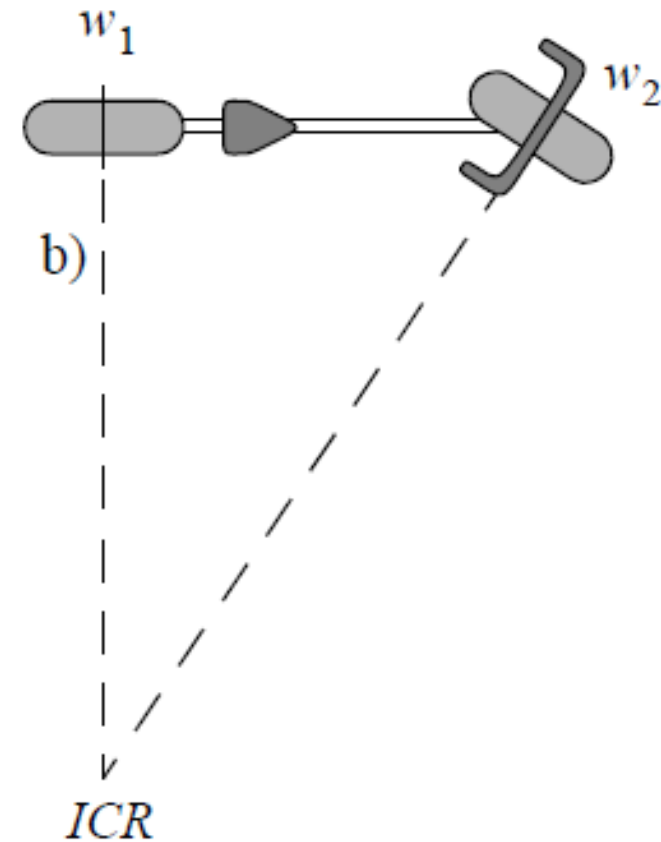
- The wheel must move instantaneously along a **circular path**:
 - With radius R
 - Whose center lies somewhere on the zero-motion line
- This center point is known as the **instantaneous center of rotation (ICR)**.
- The ICR may be located **anywhere along the zero-motion line**.
- As $R \rightarrow \infty$, the circular path becomes a **straight-line motion**.



Mobile Robot Maneuverability



(a) Four-wheel with carlike Ackerman steering.

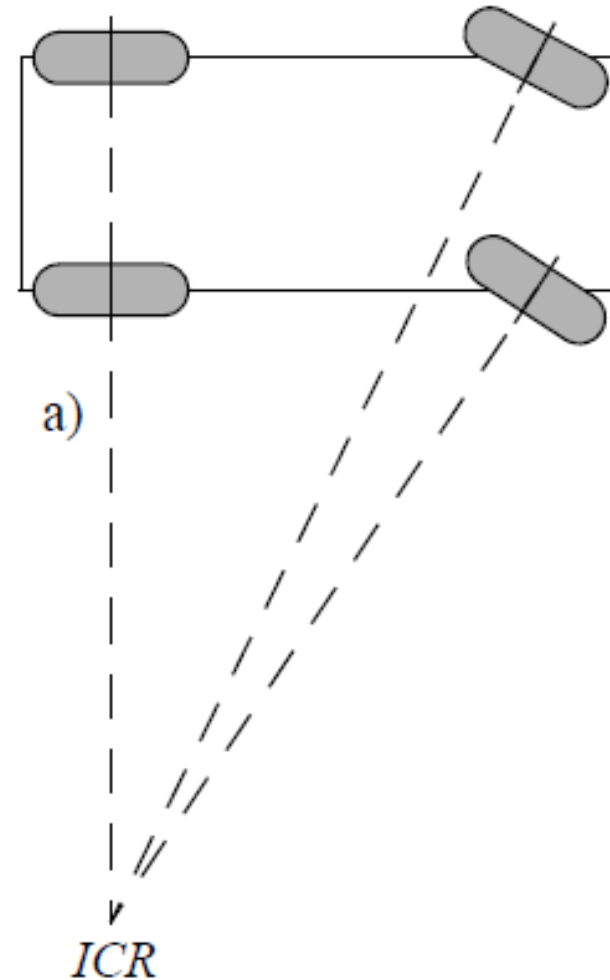


(b) Bicycle.



Mobile Robot Maneuverability

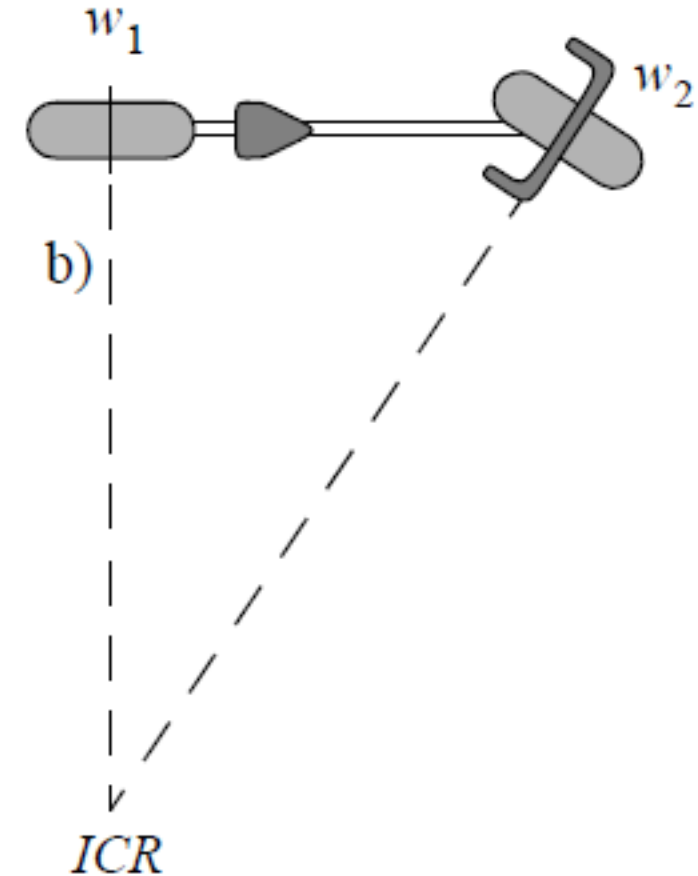
- A robot such as an **Ackermann vehicle** may have **multiple wheels**, yet it always has a **single instantaneous center of rotation (ICR)**.
- Each wheel imposes a **zero-motion line** corresponding to its sliding constraint.
- For valid motion, all zero-motion lines must **intersect at a single point**.
- This intersection uniquely determines the **ICR**, yielding a **single solution for robot motion**.
- The geometric ICR construction illustrates that:
 - Robot mobility depends on the **number of independent motion constraints** **Not** on the total number of wheels





Mobile Robot Maneuverability

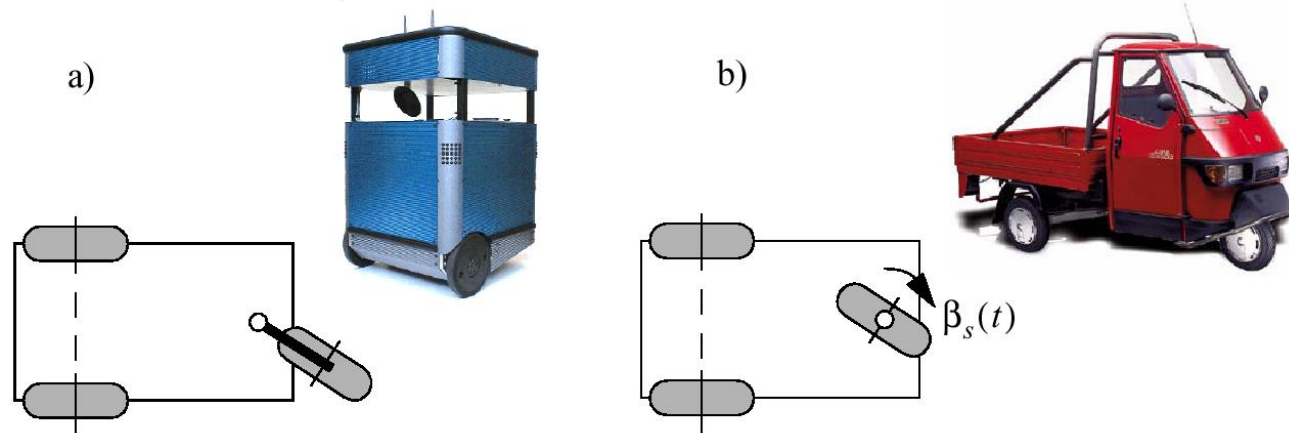
- The bicycle model consists of **two wheels**, w_1 and w_2 .
- **Each wheel imposes a kinematic constraint**, represented by a **zero-motion line**.
- When combined, the two zero-motion lines intersect at **a single point**.
- This intersection point is the **only feasible instantaneous center of rotation (ICR)**.
- The result follows because the two wheel constraints are **independent**.
- Each independent constraint **further restricts the robot's overall motion**, leaving a unique ICR.





Mobile Robot Maneuverability

- In a **differential-drive robot**, the two drive wheels are aligned along the **same horizontal axis**.
- As a result, the **instantaneous center of rotation (ICR)** is constrained to lie **along a line**, rather than at a unique point.
- Both wheels impose **identical zero-motion lines**.
- The **second wheel does not introduce an additional independent kinematic constraint**.





Mobile Robot Maneuverability

- The **Ackermann vehicle** illustrates another situation in which a wheel does **not contribute an independent kinematic constraint**.
- This vehicle has **two steerable standard wheels** (typically the front wheels).
 - The instantaneous orientation of **one steerable front wheel**, and
 - The configuration of the **fixed rear wheels**
- There exists **only a single feasible ICR**
 - The orientation of the **second steerable wheel** is **fully determined**
 - It cannot impose an additional independent constraint
- Therefore, the second steerable wheel **adds no new kinematic constraint** to the robot motion.



Mobile Robot Maneuverability

- Robot chassis kinematics are governed by the **set of independent constraints** imposed by all standard wheels.
- Wheels whose orientations are **geometrically dependent** on others:
 - Do not increase the rank of the constraint matrix
 - Do not increase mobility limitations or steering freedom



Mobile Robot Maneuverability

- The **rank of a matrix** is the maximum number of **linearly independent rows or columns**.
- The rank of $C_1(\beta_s)$ represents the **number of independent kinematic constraints**.
- A **higher rank** implies **more independent constraints**, and therefore **more restricted robot mobility**.
- As the number of independent constraints increases, the robot's allowable motions decrease.



Mobile Robot Maneuverability

- Consider a robot with **exactly one fixed standard wheel**.
- Only **standard wheels** are considered in this analysis.
- This robot may:
 - Be a **unicycle**, or
 - Include additional **Swedish wheels**
- Despite possible additional wheels, there is **only one fixed standard wheel**.
- As a result:
 - The constraint matrix has **rank one**
 - The robot has **one independent mobility constraint**



Mobile Robot Maneuverability

- In general, if

$$\text{rank}[C_{1f}] > 1,$$

- the vehicle can, at best, move **only along a circular path or a straight line**.
- This situation arises when the robot has **two or more independent constraints**.
- These constraints are introduced by **fixed standard wheels** whose **horizontal axes of rotation do not coincide**.
- Such configurations severely restrict planar motion.



Mobile Robot Maneuverability

- Robots with $\text{rank}[C_{1f}] > 1$ exhibit a **degenerate form of mobility** in the plane.
- Because of this limited mobility, **these configurations are not considered in this course.**
- Some degenerate configurations are still **useful in practice.**
- An example is the **four-wheeled slip/skid-steering system.**
- These systems perform well in certain environments, such as:
 - Loose soil
 - Sand
- However, they **violate sliding constraints.**



Mobile Robot Maneuverability

- Violating sliding constraints leads to:
 - **Reduced accuracy of dead-reckoning odometry**
 - **Significant loss in power efficiency**
- These trade-offs explain why such systems are effective in specific environments but less suitable for precise motion estimation.



Mobile Robot Maneuverability

- In general, a robot may have:
 - **Zero or more fixed standard wheels**
 - **Zero or more steerable standard wheels**
- This leads to the following possible range:
$$0 \leq \text{rank}[C_1(\beta_s)] \leq 3$$
- A rank of zero indicates **no independent kinematic constraints**.
 - There are **no fixed standard wheels**
 - There are **no steerable standard wheels**
$$N_f = 0, N_s = 0$$
- The robot therefore experiences **no motion constraints** imposed by standard wheels.



Mobile Robot Maneuverability

- Consider the opposite extreme: $\text{rank}[C_1(\beta_s)] = 3$
- This is the **maximum possible rank**, since kinematic constraints can act along at most **three degrees of freedom** in the plane (x , y , and θ).
- Consequently, there cannot be **more than three independent constraints**.
- When the rank reaches three:
 - The robot is **fully constrained in all directions**
 - No planar motion is possible
- Such a robot configuration is therefore **degenerate**, as **motion in the plane is completely prevented**.



Mobile Robot Maneuverability

- Robot's *degree of mobility* δ_m :

$$\delta_m = \dim N(C_1(\beta_s)) = 3 - \text{rank}[C_1(\beta_s)]$$

- Differential-drive chassis,

$$\delta_m = 2$$

- Bicycle chassis,

$$\delta_m = 1$$

- Omnidirectional robot,

$$\delta_m = 3$$

- δ_m defines the number of controllable DOF based on changes to wheel velocity.



Degree of Steerability

- Steering can also influence the **pose of a robot chassis**, but this influence is **indirect**.
- Changing the angle of a **steerable standard wheel** does **not immediately alter the robot pose**.
- Instead, the robot must **move** for a change in steering angle to affect its position and orientation.
- Here, we are interested in the number of **independently controllable steering parameters**.
- This quantity defines the **degree of steerability**, denoted by δ_s .

$$\delta_s = \text{rank}[C_{1s}(\beta_s)]$$



Degree of Steerability

- Recall that for mobility:
 - An increase in the rank of $C_1(\beta_s)$ implies more kinematic constraints
 - This results in a less mobile system
- In contrast, for steerability:
 - An increase in the rank of $C_{1s}(\beta_s)$ implies more steering degrees of freedom
 - This leads to greater eventual maneuverability



Degree of Steerability

- The matrix $C_1(\beta_s)$ includes $C_{1s}(\beta_s)$.
- As a result, a steered standard wheel has a dual effect:
 - Its instantaneous orientation imposes a kinematic constraint, reducing mobility
 - Its ability to change orientation enables additional motion possibilities, increasing steerability
- Thus, a steered standard wheel can simultaneously:
 - Decrease instantaneous mobility
 - Increase the set **of achievable trajectories** over time



Degree of Steerability

- The degree of steerability satisfies:

$$0 \leq \delta_s \leq 2$$

- **Case: $\delta_s = 0$** , implies that there are no steerable standard wheels.

$$N_s = 0$$

- **Case: $\delta_s = 1$ (Most Common)**
 - The case $\delta_s = 1$ is most common in robots with one or more steerable standard wheels.



Degree of Steerability

- Example: **Ordinary automobile**

- Number of fixed standard wheels: $N_f = 2$
- Number of steerable standard wheels: $N_s = 2$

- The two fixed wheels share a **common axle**, giving:

$$\text{rank}[C_{1f}] = 1$$

- The fixed wheels and **any one** steerable wheel constrain the **ICR** to lie on the line extending from the **rear axle**.

- The second steerable wheel cannot impose an additional independent constraint

$$\text{rank}[C_{1s}(\beta_s)] = 1$$

- In this configuration:

$$\delta_m = 1, \delta_s = 1$$



Degree of Steerability

- **Case: $\delta_s = 2$ — Maximum Steerability**
- The case $\delta_s = 2$ is possible only if the robot has no fixed standard wheels ($N_f = 0$).
- Two-Steer: two independently steerable standard wheels.
- This configuration is often referred to as:
 - a pseudo-bicycle, or
 - a two-steer vehicle
- Orienting the first steerable wheel constrains the instantaneous center of rotation (ICR) to lie on a line.
- Orienting the second steerable wheel further constrains the ICR to any point along that line.
 - The robot can place its ICR anywhere in the ground plane

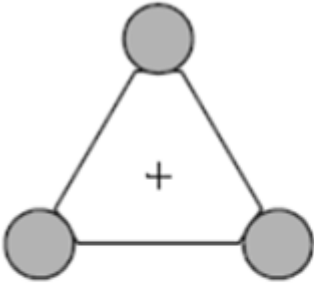
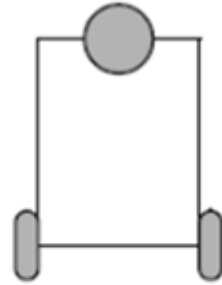
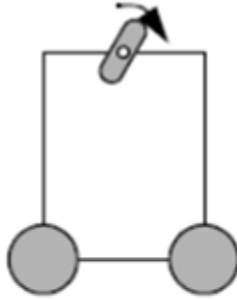
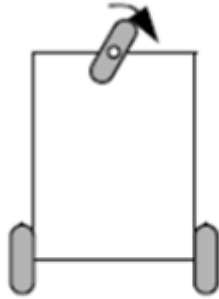
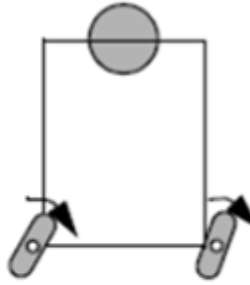


Robot maneuverability

- The overall DOF a robot can exploit are called the degree of maneuverability, denoted by δ_M .
- Maneuverability is defined in terms of mobility and steerability:
$$\delta_M = \delta_m + \delta_s$$
- δ_m : DOFs the robot can manipulate directly through wheel velocities.
- δ_s : Degrees of freedom the robot can manipulate indirectly by:
 - Changing steering configurations, and Subsequently moving the robot
- Maneuverability captures both:
 - Instantaneous motion capabilities, and
 - Additional trajectories made possible through steering adjustments over time



Robot maneuverability

				
Omnidirectional	Differential	Omni-Steer	Tricycle	Two-Steer
$\delta_m = 3$	$\delta_m = 2$	$\delta_m = 2$	$\delta_m = 1$	$\delta_m = 1$
$\delta_s = 0$	$\delta_s = 0$	$\delta_s = 1$	$\delta_s = 1$	$\delta_s = 2$
$\delta_M = 3$	$\delta_M = 2$	$\delta_M = 3$	$\delta_M = 2$	$\delta_M = 3$

The five basic types of three-wheel configurations. The spherical wheels can be replaced by castor or Swedish wheels without influencing maneuverability.



Mobile Robot Workspace

- The sliding constraint for a **fixed standard wheel** is given by:
$$[\cos(\alpha + \beta) \quad \sin(\alpha + \beta) \quad l\sin\beta]R(\theta) \dot{\xi}_I = 0$$
- This constraint is expressed in terms of the **robot motion** $\dot{\xi}$, not the pose ξ .
- The purpose of the constraint is to enforce **zero velocity perpendicular to the wheel plane**.



Mobile Robot Workspace

- **Non-Holonomic Nature of the sliding Constraint**
 - The sliding constraint depends explicitly on robot motion.
 - It cannot be integrated into a constraint on position alone.
 - Therefore, the sliding constraint is non-integrable.
 - The fixed standard wheel sliding constraint is a non-holonomic constraint.
- **Example: Bicycle Configuration**
 - One fixed standard wheel
 - One steerable standard wheel
- Because the fixed wheel's sliding constraint is active:
 - The robot is subject to a non-holonomic constraint
- Thus, the bicycle is a non-holonomic robot.



Mobile Robot Workspace

- Consider a bicycle whose steering mechanism is locked.
- The system then becomes equivalent to two fixed standard wheels with separate but parallel axes.
- For this configuration, the degree of maneuverability is:

$$\delta_M = 1$$

- **Is the Locked Bicycle Non-Holonomic?**
- At first glance, the presence of sliding and rolling constraints may suggest non-holonomic behavior.
- However, **the locked bicycle is actually holonomic.**
- This is because its constraints can be fully expressed as constraints on position, rather than on motion.



Mobile Robot Workspace

- **In Locked Bicycle**, the **sliding constraints of both wheels** can be replaced by an **equally complete set of constraints on the robot pose**.
- These pose-level constraints fully characterize the robot's allowable configurations.
- The **two non-holonomic sliding constraints** associated with the wheels are **eliminated**.
- The system is no longer constrained by motion-level (velocity-dependent) restrictions.



Mobile Robot Workspace

- The locked bicycle is an example of the **first type of holonomic robot**:
 - Constraints **do exist**
 - All constraints are **holonomic kinematic constraints**
- Even though mobility is limited, the robot's motion is:
 - Fully describable by **position constraints**
 - Free of any **non-integrable motion constraints**



Mobile Robot Workspace

- The second type of holonomic robot occurs when **no kinematic constraints exist**.
- This corresponds to:

$$N_f = 0, N_s = 0$$

- With no kinematic constraints:
 - There are **no non-holonomic constraints**
 - The robot is **always holonomic**
- All holonomic robots with $\delta_M = 3$ fall into this category.



Mobile Robot Workspace

- In mobile robotics, useful Robot must generally reach any pose in the plane.
- Therefore, the workspace must have **dimensionality 3**:
$$\text{DOF} = 3(x, y, \theta)$$
- Beyond reachability, important capabilities include:
 - Maneuvering around obstacles **without changing orientation**
 - Tracking a target while following an **arbitrary path**



Mobile Robot Workspace

- The most relevant form of holonomy in mobile robotics satisfies:
$$\text{DDOF} = \text{DOF} = 3$$
- Robots that meet this condition are defined as **omnidirectional**.
- An **omnidirectional robot** is therefore:
 - **Holonomic**
 - Capable of independent control of position and orientation in the plane



Mobile Robot Workspace

- Maneuverability is equivalent to the vehicle's DOFs
- But what is the degree of vehicle's freedom in its environment?
 - Car example
- Workspace
 - how the vehicle is able to move between different configuration in its workspace?
- The robot's independently achievable velocities
 - δ_m = differentiable degrees of freedom (DDOF)
 - Bicycle: $\delta_M = \delta_m + \delta_s = 1 + 1$ $DDOF = 1$; $DOF = 3$
 - Omnibot: $\delta_M = \delta_m + \delta_s = 3 + 0$ $DDOF = 3$; $DOF = 3$



Mobile Robot Workspace

- DOF degrees of freedom:
 - Robots' ability to achieve various poses
- DDOF differentiable degrees of freedom:
 - Robots' ability to achieve various path

$$DDOF \leq \delta_m \leq DOF$$

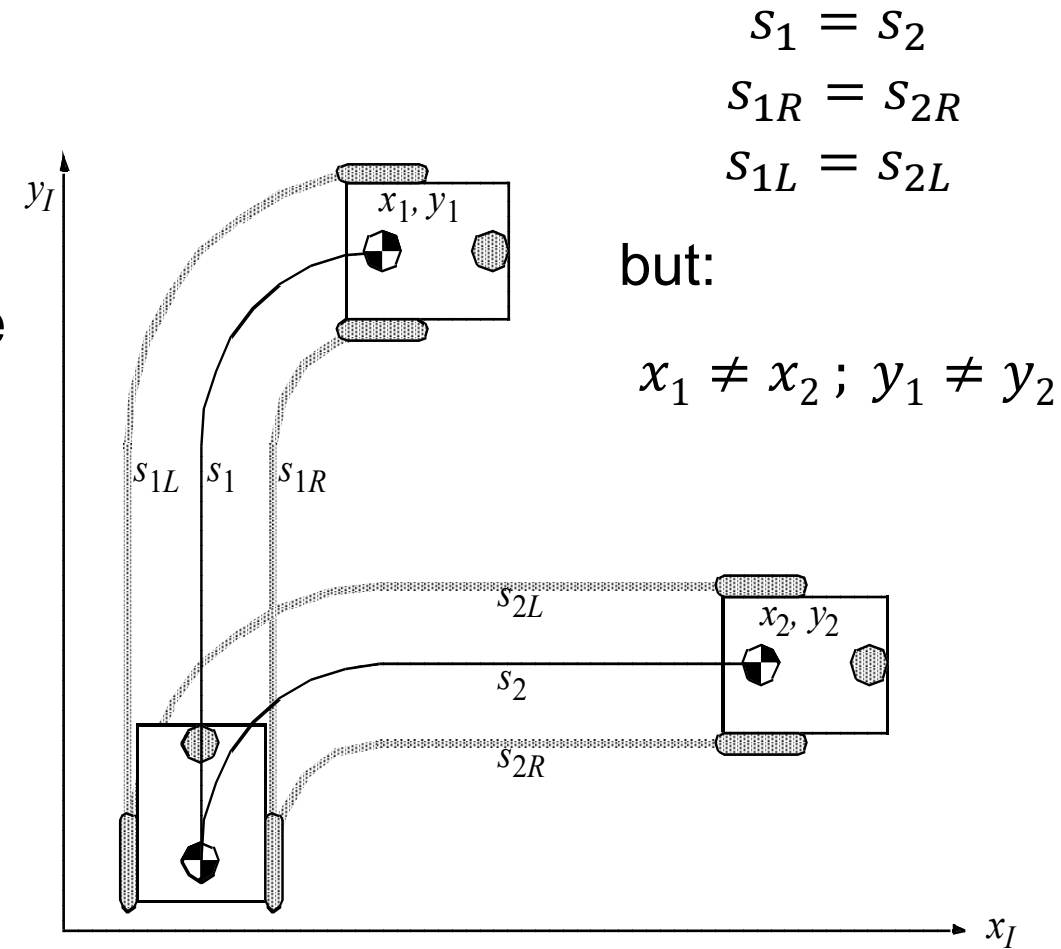
- **Holonomic Robots**

- A holonomic kinematic constraint can be expressed as an explicit function of position variables only
- A non-holonomic constraint requires a different relationship, such as the derivative of a position variable
- Fixed and steered standard wheels impose non-holonomic constraints



Non-Holonomic Robots

- In non-holonomic systems, the governing differential equations are not integrable to yield the final position directly.
- Knowledge of the total distance traveled by each wheel is not sufficient to determine the robot's final pose.
- To compute the final configuration of the robot, one must also know:
 - How the motion was executed over time
 - That is, the time history of the control inputs
- The final position depends on the path taken, not just the start and end values of wheel displacements.





Path / Trajectory Considerations

- We care not only about a robot's ability to **reach a desired final configuration**, but also about **how it gets there**.
- An important consideration is the robot's ability to **follow paths**.
- Ideally, a robot should be able to **trace any path** in its workspace of poses.
- **Omnidirectional robots** can follow arbitrary paths because they are:
 - **Holonomic**
 - Operating in a **three-dimensional workspace** (x, y, θ)
- This always allows independent control of position and orientation.



Path / Trajectory Considerations

- Omnidirectional robots must use **unconstrained wheels**, such as:
 - Swedish wheels
 - Castor wheels
 - Spherical wheels
- These wheel types typically:
 - Limit achievable **ground clearance**
 - Make the use of **suspensions** difficult
- Although omnidirectional robots are **very powerful from a path-space perspective**:
 - Their mechanical design is **complex**
 - Fabrication is often **expensive**
- These trade-offs must be considered when selecting a chassis for real-world applications.



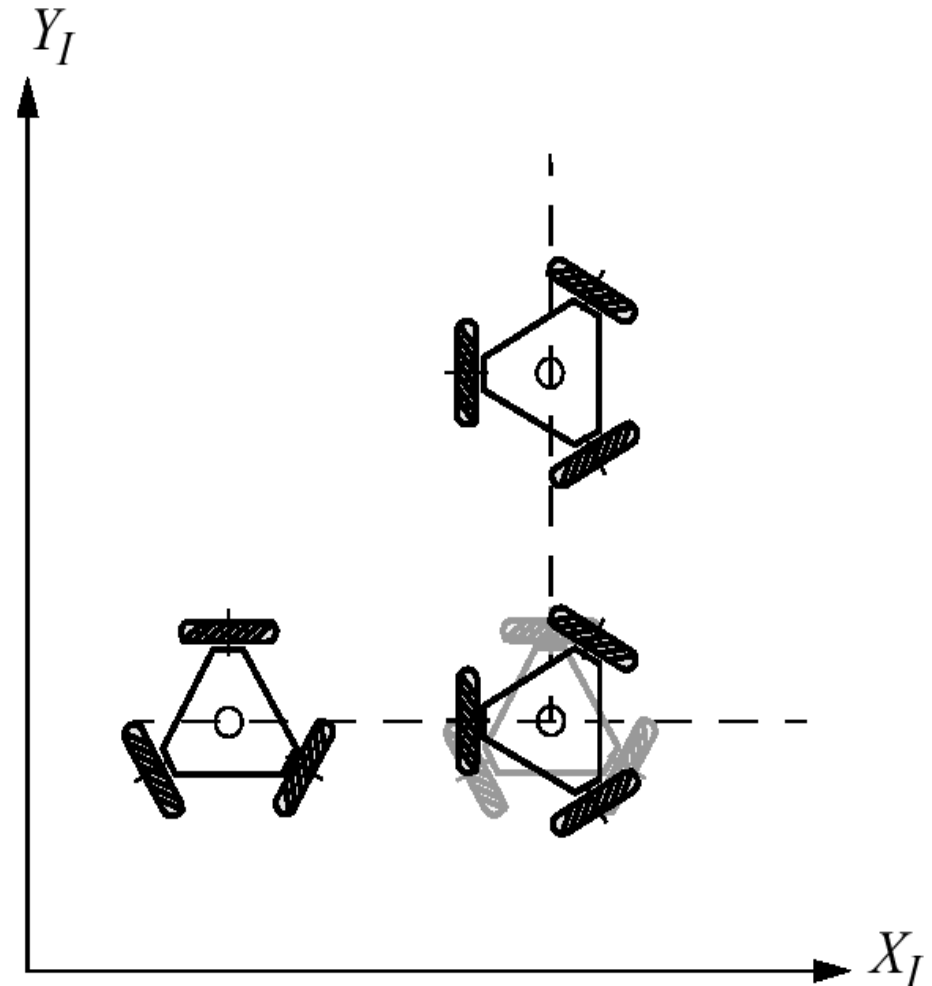
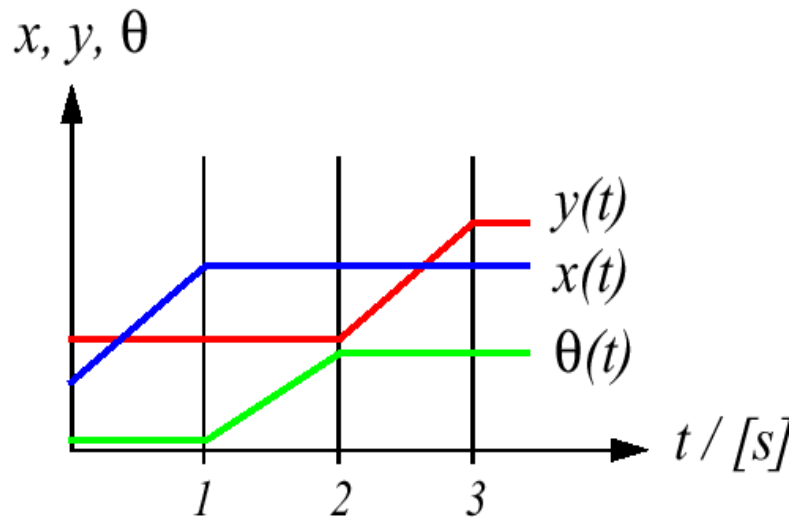
Path / Trajectory Considerations

- Non-holonomic constraints can **significantly improve motion stability** in mobile robots.
- A **car-like robot** is subject to **kinematic (sliding) constraints passively counteract lateral forces** during curved motion.
 - Stability is improved
 - Demands on motor torque are reduced
- This passive stabilization is an important practical advantage of **non-holonomic designs**.
- **Omnidirectional vehicle** driving at high speed along a curve
 - The vehicle experiences a **centripetal (lateral) force**.
 - This lateral force pushes the vehicle **outward from the curve**.
 - This force must be **actively counteracted by wheel motor torques**.
 - In the event of a **motor or control failure**, the vehicle may be **thrown out of the curve**.



Path / Trajectory Considerations

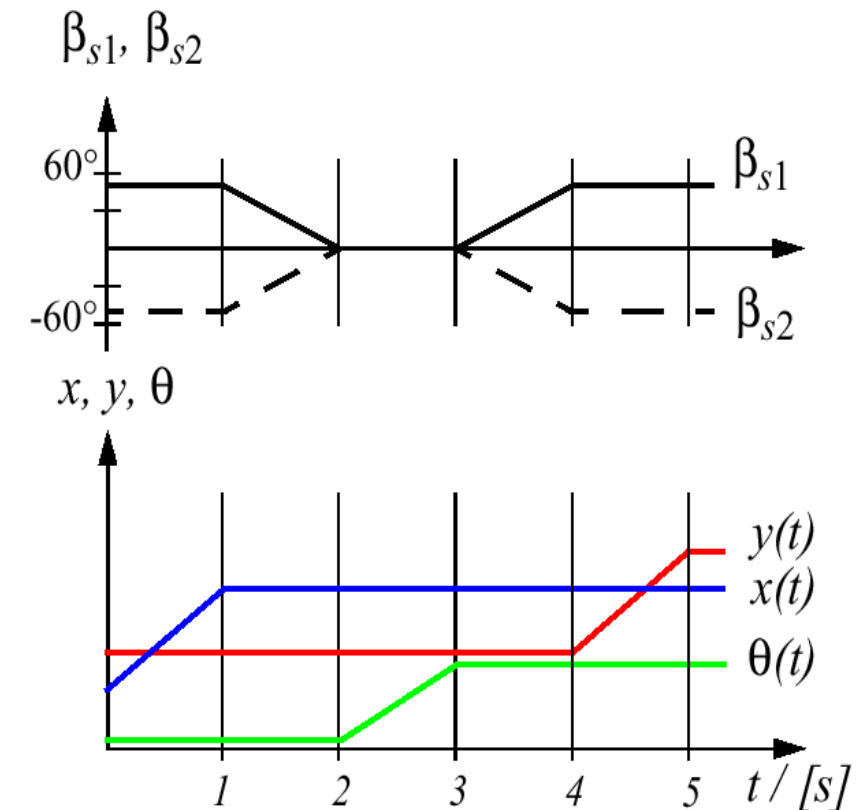
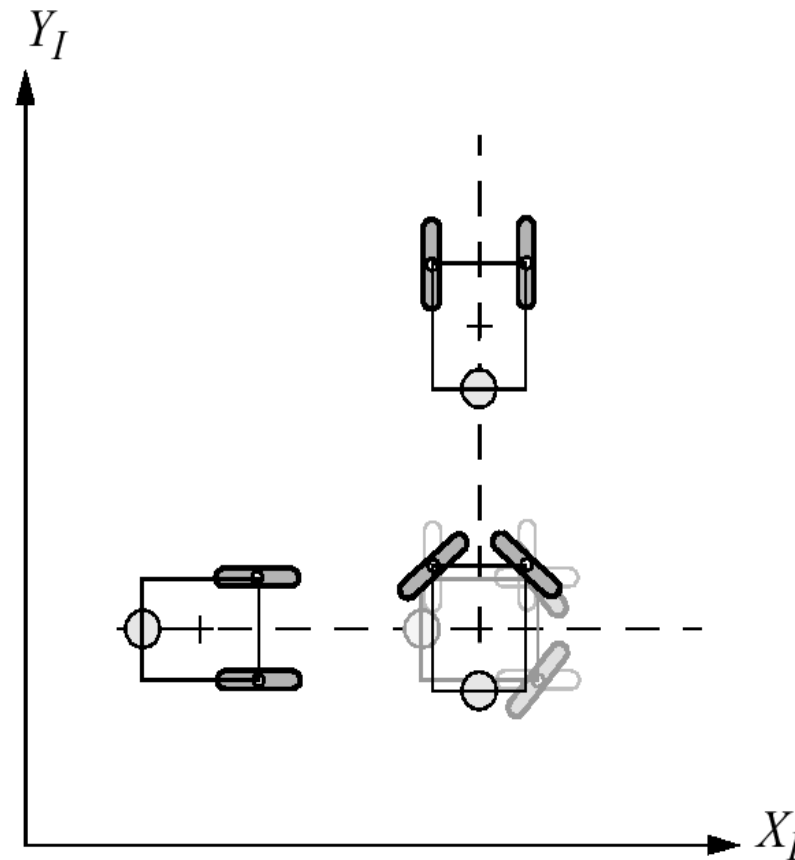
- Example of robot trajectory with omnidirectional robot: move for 1 second with constant speed of 1 m/s along axis X_I ; change orientation counterclockwise 90 degree, in 1 second; move for 1 second with constant speed of 1 m/s along axis Y_I .





Path / Trajectory Considerations

Two steered wheels: move for 1 second with constant speed of 1 m/s along axis X_I ; rotate steered wheels -50 / 50 degree respectively; change orientation counterclockwise 90 degree in 1 second; rotate steered wheels 50 / -50 degree respectively; move for 1 second with constant speed of 1 m/s along axis Y_I .





Path / Trajectory Considerations

- Cars cannot move sideways; it must follow kinematic constraints (limited steering angle, turning radius, forward/reverse only).
- In tight parking spaces, these constraints force multi point maneuvers to achieve lateral repositioning and final alignment.
- Many autonomous parking systems use search/planning (e.g., Hybrid A* and variants) to generate feasible, collision free trajectories under non holonomic constraints.
- Sampling planners (e.g., Rapidly Exploring Random Tree RRT family) explore many feasible paths and can improve path quality with more samples, but still operate within explicit vehicle constraints.



Path / Trajectory Considerations

- RL can learn by trial-and-error to maximize a reward (success, clearance, smoothness), so it is not limited to copying human habits.
- This can yield unexpected maneuver sequences (approach angles, micro-corrections, reversal patterns) that humans rarely use.
- Practical systems often combine learning with structured methods (planning and/or imitation) to keep behaviour safe, feasible, and robust.



Any Questions